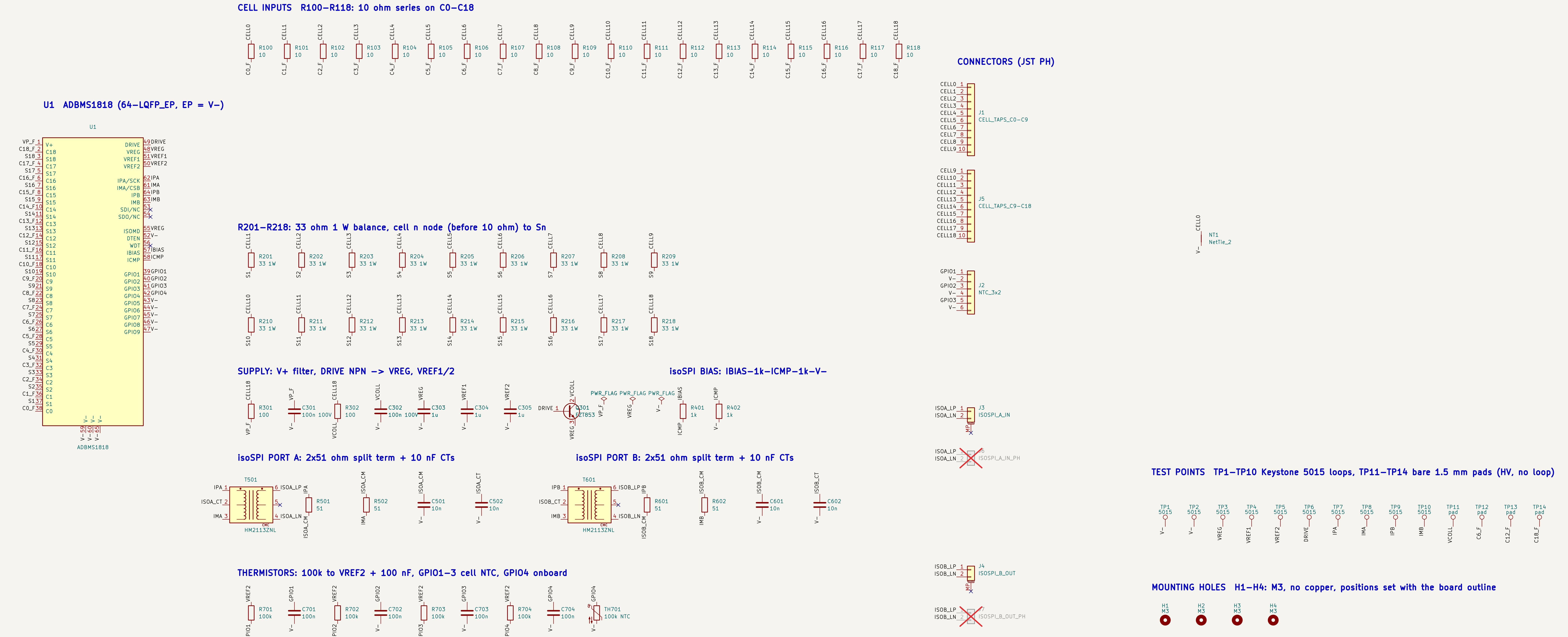
Wired with net labels: labels with the same name are one net. 18S Lean: no fuses, clamps or C–pin caps; generated from slave–18s by make_lean.py, do not edit by hand. C0 sense (CELL0, J1.1) and V– meet only at net tie NT1, which replaces F120.
ADBMS1818 Figure 112, isoSPI filtering per Fig. 104(B). See ../docs/slave–18s–design–basis.md.


CELL INPUTS R100–R118: 10 ohm series on C0–C18
R100 10 … R118 10
CELL0 … CELL18
C0_F … C18_F
U1 ADBMS1818 (64–LQFP_EP, EP = V–)
U1
ADBMS1818
V+ C18 S18 C17 S17 C16 S16 C15 S15 C14 S14 C13 S13 C12 S12 C11 S11 C10 S10 C9 S9 C8 S8 C7 S7 C6 S6 C5 S5 C4 S4 C3 S3 C2 S2 C1 S1 C0
DRIVE VREG VREF1 VREF2 IPA/SCK IMA/CSB IPB IMB SDI/NC SDO/NC ISOMD DTEN WDT IBIAS ICMP GPIO1 GPIO2 GPIO3 GPIO4 GPIO5 GPIO6 GPIO7 GPIO8 GPIO9
R201–R218: 33 ohm 1 W balance, cell n node (before 10 ohm) to Sn
R201 33 1W … R218 33 1W
CONNECTORS (JST PH)
J1 CELL_TAPS_C0–C9
J5 CELL_TAPS_C9–C18
J2 NTC_3x2
NT1 NetTie_2
SUPPLY: V+ filter, DRIVE NPN –> VREG, VREF1/2
R301 100
C301 100n 100V
R302 100
C302 100n 100V
C303 1u
C304 1u
C305 1u
Q301 FZT853
PWR_FLAG PWR_FLAG PWR_FLAG
isoSPI BIAS: IBIAS–1k–ICMP–1k–V–
R401 1k
R402 1k
J3 ISOSPI_A_IN
J6 ISOSPI_A_IN_PH
J4 ISOSPI_B_OUT
J7 ISOSPI_B_OUT_PH
isoSPI PORT A: 2x51 ohm split term + 10 nF CTs
T501 HM2113ZNL
R501 51
R502 51
C501 10n
C502 10n
isoSPI PORT B: 2x51 ohm split term + 10 nF CTs
T601 HM2113ZNL
R601 51
R602 51
C601 10n
C602 10n
TEST POINTS TP1–TP10 Keystone 5015 loops, TP11–TP14 bare 1.5 mm pads (HV, no loop)
TP1 5015 … TP10 5015
TP11 pad TP12 pad TP13 pad TP14 pad
THERMISTORS: 100k to VREF2 + 100 nF, GPIO1–3 cell NTC, GPIO4 onboard
R701 100k
C701 100n
R702 100k
C702 100n
R703 100k
C703 100n
R704 100k
C704 100n
TH701 100k NTC
MOUNTING HOLES H1–H4: M3, no copper, positions set with the board outline
H1 M3
H2 M3
H3 M3
H4 M3

Not for manufacture
Baseline: ADBMS1818 datasheet Figure 112 typical application

Sheet: /
File: adbms1818–slave–18s–lean.kicad_sch

Title: ADBMS1818 18S slave module – Lean (no hot–plug protection)

Size: User
Date: 2026–11–06
Rev: 0–draft
KiCad E.D.A. 10.0.6
Id: 1/1